Modularity of Combinatorial Rank Generating Functions

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Motivation for Modular Forms

Modular forms and their generalizations play a central role in number theory. They have connections to Algebraic geometry, Combinatorics, Mathematical Physics, Representation Theory, Topology, and more!

Classical Modular Forms Set Up

The group

$\mathrm{SL}_2(\mathbb{Z}) = \{\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mid \mathrm{a, b, c, d} \in \mathbb{Z}, \det\gamma = 1\} = \langle \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \rangle = \langle \mathrm{T, S} \rangle$

acts on the set

$$\mathbb{H} = \{\tau = x + iy \mid x, y \in \mathbb{R}, y > 0\}$$

via

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\tau = \frac{a\tau + b}{c\tau + d}.$$

Observe that $T\tau = \tau + 1$ and $S\tau = -\frac{1}{\tau}$.

Classical Modular Forms Set Up

This action extends to include $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{i\infty\}$. For $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, $\Gamma\backslash\mathbb{H}$ is noncompact, but letting $\mathbb{H}^* = \mathbb{H} \cup \mathbb{P}^1(\mathbb{Q})$, then $\Gamma\backslash\mathbb{H}^*$ is a compact Riemann surface.

We call elements of $\Gamma\backslash\mathbb{P}^1(\mathbb{Q})$ *cusps* of Γ.

Definition of Modular Form

A classical modular form is a function $f : \mathbb{H} \to \mathbb{C}$ with respect to a group $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, such that

1. f satisfies an analyticity condition on $\mathbb{H}$ and at the cusps of Γ.
2. f satisfies a modular transformation formula. Namely, for all $\tau \in \mathbb{H}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$,

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = \epsilon_\gamma (c\tau + d)^k f(\tau).$$

Here $k \in \frac{1}{2}\mathbb{Z}$ is called the *weight* of f, and ϵ_γ is called the *multiplier* and is a root of unity.

Let $q = e^{2\pi i \tau}$. Then f has a Fourier expansion $f(\tau) = \sum_{n \geqslant 0} a(n) q^n$, with $a(n) \in \mathbb{C}$.

Generalizations of Modular Forms

- weakly holomorphic modular form
- meromorphic modular form
- Maass form
- vector valued modular form
- mock modular form
- quantum modular form
- $\vdots$

Example: Dedekind's η-function

Dedekind's η-function is defined by

$$\eta(\tau) := q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1 - q^n).$$

This function is one of the most well-loved modular forms!

Modularity properties of η

For all $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $\tau \in \mathbb{H}$, we have that

$$\eta\left(\gamma\tau\right) = \psi\left(\gamma\right)\left(c\tau + d\right)^{\frac{1}{2}}\eta(\tau), \tag{1}$$

where $\psi\left(\gamma\right)$ is a 24th root of unity, which can be given explicitly in terms of Dedekind sums. In particular,

$$\eta\left(\frac{-1}{\tau}\right) = e^{-\frac{\pi i}{4}}\tau^{\frac{1}{2}}\,\eta(\tau)$$

$$\eta(\tau + 1) = e^{\frac{\pi i}{12}}\eta(\tau).$$

Ramanujan's Enigmatic Deathbed Letter

- Sent to Hardy in 1920
- Lists 17 q-series that he calls "mock" theta functions
- He says that unlike "false" theta functions, they enter into mathematics as beautifully as theta functions...
- He examined the asymptotic difference between mock theta functions and theta functions at "radial limits"
- No one knew what he meant before he died!
- The modular properties of mock theta functions remained unknown for nearly a century...

Mock Modular Forms

- In his 2002 PhD thesis, Zwegers showed that these mock theta functions could be completed to form a weight 1/2 harmonic Maass form, which is modular but no longer holomorphic.
- Harmonic Maass forms, in addition to being modular, also satisfy bounded growth conditions and are annihilated by a weighted Laplacian:

$$\Delta_k = -y^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + iky \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

- Zagier defined *mock modular form* to mean the holomorphic part of a harmonic Maass form.

Integer Partitions

A partition of a positive integer n is a non-increasing sequence of positive integers that sum to n.

For example, the following are all the partitions of 4:

$$
\begin{aligned}
&4\\
&3+1\\
&2+2\\
&2+1+1\\
&1+1+1+1.
\end{aligned}
$$

Partition Function $p(n)$

$$p(n) = \text{ the number of partitions of } n.$$

Some values are:

n	0	1	2	3	4	5	6	7	$\cdots$	200
$p(n)$	1	1	2	3	5	7	11	15	$\cdots$	$3,972,999,029,388$

Generating Function for $p(n)$

$$\sum_{n=1}^{\infty} p(n)q^n = \prod_{n=1}^{\infty} \frac{1}{1-q^n} = q^{\frac{1}{24}}\eta(\tau)^{-1},$$

when $q = e^{2\pi i\tau}$, so is connected to modular forms!

To see this, use geometric series for $|q| < 1$:

$$\prod_{n=1}^{\infty} \frac{1}{1-q^n} = \prod_{n=1}^{\infty} (1 + q^n + q^{n+n} + q^{n+n+n} + \cdots).$$

The connection between modular forms and partition functions is much deeper than one might initially expect...

Rank of a Partition

The rank of a partition is its largest part minus its number of parts. For example, the rank of $2+1+1$ is -1.

Let $N(m,n) := \#\{\text{partitions of } n \text{ with rank equal to } m\}$. Then the two variable generating function for $N(m,n)$ can be expressed as

$$\sum_{m=-\infty}^{\infty}\sum_{n=0}^{\infty} N(m,n)x^m q^n = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(xq;q)_n(x^{-1}q;q)_n} =: R_1(x;q),$$

where $N(m,0) = \delta_{m0}$, and

$$(a;q)_n := \prod_{j=1}^{n}(1-aq^{j-1}).$$

Specializing $R_1(x;q)$ at roots of unity x

- For $x = 1$,

$$R_1(1;q) = 1 + \sum_{n=1}^{\infty} p(n)q^n = q^{\frac{1}{24}}\eta^{-1}(\tau),$$

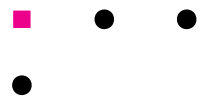
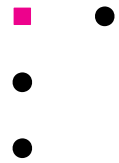
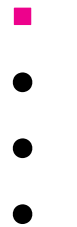
  so $q^{-\frac{1}{24}}R_1(1;q)$ is a modular form.
- For $x = -1$,

$$q^{-\frac{1}{24}}R_1(-1;q) = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(-q;q)_n^2} =: f(q),$$

  which is one of Ramanujan's original third order mock theta functions, a mock modular form.
- In 2010, Bringmann and Ono used the theory of harmonic Maass forms to prove that for general roots of unity $x \neq 1$, $R_1(x;q)$ is also a mock modular form.

Generalization: Durfee Symbols

Andrews defined Durfee symbols, which provide an alternate representation of a partition. For example,

4	$3+1$	$2+2$	$2+1+1$	$1+1+1+1$
				
$\begin{pmatrix} 1 & 1 & 1 \\ & & \end{pmatrix}_1$	$\begin{pmatrix} 1 & 1 \\ 1 & \end{pmatrix}_1$	$\begin{pmatrix} & \end{pmatrix}_2$	$\begin{pmatrix} 1 & \\ 1 & 1 \end{pmatrix}_1$	$\begin{pmatrix} & & \\ 1 & 1 & 1 \end{pmatrix}_1$

The rank of a Durfee symbol is the length of the partition in the top row, minus the length of the partition in the bottom row. This gives Dyson's rank of a partition.

n-marked Durfee Symbols

Andrews generalized this to n-marked Durfee symbols using n copies of the integers. Here is a 3-marked Durfee symbol of 55.

$$\left(\begin{array}{cc|ccc|c} 4_3 & 4_3 & 3_2 & 3_2 & 2_2 & 2_1 \\ & 5_3 & & 3_2 & 2_2 & 2_1 \end{array}\right)_5 =: \left(\begin{array}{c|c|c} \alpha^3 & \alpha^2 & \alpha^1 \\ \beta^3 & \beta^2 & \beta^1 \end{array}\right)_5$$

Each n-marked Durfee symbol has n ranks. The jth rank is defined by

$$\begin{aligned} &\mathrm{length}(\alpha^{\mathrm{j}}) - \mathrm{length}(\beta^{\mathrm{j}}) - 1,\ \mathrm{j} < \mathrm{n}, \\ &\mathrm{length}(\alpha^{\mathrm{n}}) - \mathrm{length}(\beta^{\mathrm{n}}),\ \mathrm{j} = \mathrm{n}. \end{aligned}$$

In the above example, the 3rd rank is 1, 2nd rank is 0, 1st rank is -1.

Generating function for $\mathcal{D}_n(m_1, m_2, \ldots, m_n; r)$

Let $D_m(m_1, \ldots, m_n; r)$ count the number of n-marked Durfee symbols of r with ith rank equal to m_i. Andrews showed that the $(n+1)$-variable rank generating function for $\mathcal{D}_n(m_1, m_2, \ldots, m_n; r)$ may be expressed similarly to that of $R_1(x; q)$.

Theorem (Andrews)

For $n \geqslant 1$,

$$\sum_{m_1, m_2, \ldots, m_n = -\infty}^{\infty} \sum_{r=0}^{\infty} \mathcal{D}_n(m_1, m_2, \ldots, m_n; r) x_1^{m_1} x_2^{m_2} \cdots x_n^{m_n} q^r = R_n(\boldsymbol{x}; q).$$

Definition of $R_n(\boldsymbol{x};q)$

For $n \geqslant 2$,

$$R_n(\boldsymbol{x};q) := \sum_{\substack{m_1>0\\ m_2,\ldots,m_n\geqslant 0}} \frac{q^{(m_1+m_2+\cdots+m_n)^2+(m_1+\cdots+m_{n-1})+(m_1+\cdots+m_{n-2})+\cdots+m_1}}{(x_1q;q)_{m_1}\left(\frac{q}{x_1};q\right)_{m_1}(x_2q^{m_1};q)_{m_2+1}\left(\frac{q^{m_1}}{x_2};q\right)_{m_2+1}\cdots(x_nq^{m_1+\ldots+m_{n-1}};q)_{m_n+1}\left(\frac{q^{m_1+\ldots+m_{n-1}}}{x_n};q\right)_{m_n+1}}.$$

Specializing $R_n(\boldsymbol{x}; q)$ at roots of unity x

- For $n = 2$, $\boldsymbol{x} = (1, 1)$, in 2008 Bringmann showed that $R_2(1, 1; q)$ is a quasimock modular form. This means it is a linear combination of derivatives of mock modular forms.
- For $n \geqslant 2$, $\boldsymbol{x} = (1, \ldots, 1)$, in 2009, Bringmann, Garvan, and Mahlburg showed that $R_n(1, \ldots, 1; q)$ is a quasimock modular form.
- In 2013, Folsom and Kimport show that for more general vectors of roots of unity $\boldsymbol{x}$, $R_n(\boldsymbol{x}; q)$ is a linear combination of mixed mock and quasimock modular forms.
- In 2019, Folsom, Kimport, Jang, and Swisher proved that for the same vectors of roots of unity as above, $R_n(\boldsymbol{x}; q)$ has quantum modular form properties...

Quantum Modular Forms

- In 2010, Zagier defined a new type of modular object which he called *quantum modular forms*, which are complex functions defined on $\mathbb{Q}$ (or a subset) such that the "errors to modularity"

$$f(x) - (cx + d)^{-k} f(\gamma x)$$

are analytic on some subset of $\mathbb{R}$ with the usual real topology.

Motivation for Definition

- One shouldn't define a quantum modular form to be analytic on the cusps, since the cusps have the discrete topology so requiring continuity or analyticity is vacuous. (Not restrictive enough)
- One shouldn't define a quantum modular form to satisfy

$$f(\gamma x) = (cx + d)^k f(x)$$

for all $x \in \mathbb{Q}$ and $\gamma \in \Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, since there will be only finitely many orbits. (Too restrictive)

Broader Context

Examples of quantum modular forms have arisen in many interesting areas, including

- quadratic forms (Zagier)
- quantum invariants of 3-manifolds (Zagier)
- n-colored Jones polynomials of knots (Zagier, Hickami and Lovejoy)
- Unimodal sequences (Bryson, Ono, Pitman, Rhoades)
- meromorphic Jacobi forms (Bringmann, Rolen, Zwegers)

Definition of Quantum Modular Form

More precisely,

- There is a subset $S \subseteq \mathbb{Q}$ and group $G \subseteq \mathrm{SL}_2(\mathbb{Z})$ such that both $f(x)$ and $f(Mx)$ exist (are non-singular) for all $x \in S$ and $M \in G$. Here, S is called a *quantum set* for f with respect to the group G.
- There is a weight $k \in \frac{1}{2}\mathbb{Z}$ such that for all $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in G$, $x \in S \backslash \{-\frac{d}{c}\}$, the functions

$$h_{f,\gamma}(x) := f(x) - \epsilon(\gamma)(cx+d)^{-k} f\left(\frac{ax+b}{cx+d}\right)$$

  are analytic on (some subset of) $\mathbb{R}$.
  Here the $\epsilon(\gamma)$ are appropriate complex numbers as with $\eta(\tau)$.

Quantum modularity and mock modular forms

Quantum modular forms relate to Ramanujan's original definition of a mock theta function!

The asymptotic differences between mock theta functions and modular theta functions at "radial limits" are now known to be specific values of quantum modular forms.

Example 1: Kontsevich-Zagier function

Let $q = e^{2\pi i x}$, where $x \in \mathbb{Q}$. Then,

$$F(x) := \sum_{n \geqslant 0} (1-q)(1-q^2)\cdots(1-q^n) = \sum_{n \geqslant 0} (q;q)_n.$$

Note that F doesn't converge on any open subset of $\mathbb{C}$. However, F is defined for any $x \in \mathbb{Q}$, since then the sum truncates!

In 2010, Zagier proved that $e^{\frac{\pi i x}{24}} F(x)$ is a weight $3/2$ quantum modular form.

Example 2: Unimodal rank generating function

A strongly unimodal sequence of size n is a list of positive integers $a_1, \ldots, a_s$ such that $\sum_i a_i = n$ and

$$a_1 < a_2 < \cdots < a_k > a_{k+1} > \cdots > a_s.$$

We write $u(n)$ to count the number of strongly unimodal sequences of size n.

For example, the following are all the strongly unimodal sequences of size 4:

$$\begin{aligned} &4 \\ &1, 3 \\ &3, 1 \\ &1, 2, 1 \end{aligned}$$

so we have that $u(4) = 4$.

Example 2: Unimodal rank generating function

The rank of a strongly unimodal sequence of size n is the number of terms right of the peak minus the number of terms left of the peak.

We write $u(m, n)$ to count the number of strongly unimodal sequences of size n with rank m.
For example,

unimodal sequence	rank
4	0
$1, 3$	-1
$3, 1$	1
$1, 2, 1$	0

so we have $u(0, 4) = 2$.

Example 2: Unimodal rank generating function

The generating function for $u(m,n)$ is given by

$$U(x;q) = \sum_{m\in\mathbb{Z}} \sum_{n\geqslant 0} u(m,n) x^m q^n = \sum_{n\geqslant 0} (-xq;q)_n (-xq^{-1};q)_n q^{n+1}.$$

Duality between $U(-1;q)$ and $F(x)$

Theorem (Bryson, Ono, Pitman, Rhoades (2012))

1. *When q is a root of unity,*

$$U(-1;q) = F(q^{-1}).$$

2. *Moreover, $e^{\frac{-\pi i x}{12}} U(-1; e^{2\pi i x})$ is a weight $3/2$ quantum modular form with respect to $\mathrm{SL}_2(\mathbb{Z})$.*

Duality generalizations

There have been many interesting generalizations, including:

- Hickami, Lovejoy (2015): Prove a similar type of duality between a generalized U function and a generalized family of quantum modular forms coming from colored Jones polynomials of torus knots at roots of unity.
- Folsom, Ki, Truong, Yang (2017): Generalized duality and quantum modularity for 2-variable families.
- Folsom (2019): Extends work of Hickami, Lovejoy to 2-variable quantum Jacobi forms
- Lovejoy, Osburn (2019): Further duality theorems arising from colored Jones polynomials of double twist knots.

Our Project

Our project is to generalize the 2-variable rank generating function for strongly unimodal sequences to a multivariable multi-rank generating function like Andrews' generating function for n-marked Durfee symbols.

1. Generalize $U(x;q)$ to a multiple variable version $U_n(\vec{x};q)$.
2. Plug in vectors of roots of unity for $\vec{x}$ and study potential modularity properties.
3. Plug in vectors of roots of unity for $\vec{x}$ and study potential quantum modularity properties.
4. Try to generalize the duality theorem of Bryson, Ono, Pitman, Rhoades to a multivariable setting.